MSG No. 204.31 $Im\bar{3}1'$ [Type II, cubic]

Table 1: Wyckoff site: 2a, site symmetry: m-3.1'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,
		73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96]

Table 2: Wyckoff site: 6b, site symmetry: mmm..1'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,13,14,15,16,49,50,51,52,61,62,63,64]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,8,10,12,17,20,22,24,53,56,58,60,65,68,70,72]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[6,7,9,11,18,19,21,23,54,55,57,59,66,67,69,71]
4	$\left[\frac{1}{2}, 0, 0\right]$	[25,26,27,28,37,38,39,40,73,74,75,76,85,86,87,88]
5	$\left[0, \frac{1}{2}, 0\right]$	[29,32,34,36,41,44,46,48,77,80,82,84,89,92,94,96]
6	$\left[0, 0, \frac{1}{2}\right]$	[30,31,33,35,42,43,45,47,78,79,81,83,90,91,93,95]

Table 3: Wyckoff site: 8c, site symmetry: .-3'.1'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,5,6,37,41,42,49,53,54,85,89,90]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[2,10,11,38,46,47,50,58,59,86,94,95]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,7,12,39,43,48,51,55,60,87,91,96]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,8,9,40,44,45,52,56,57,88,92,93]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[13,17,18,25,29,30,61,65,66,73,77,78]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[14,22,23,26,34,35,62,70,71,74,82,83]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[15,19,24,27,31,36,63,67,72,75,79,84]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[16,20,21,28,32,33,64,68,69,76,80,81]

Table 4: Wyckoff site: 12d, site symmetry: 2mm..1'

No.	position	mapping
1	$[x, 0, 0]$	[1,2,15,16,49,50,63,64]
2	$[-x, 0, 0]$	[3,4,13,14,51,52,61,62]
3	$[0, x, 0]$	[5,12,20,22,53,60,68,70]
4	$[0, 0, x]$	[6,9,19,23,54,57,67,71]
5	$[0, 0, -x]$	[7,11,18,21,55,59,66,69]
6	$[0, -x, 0]$	[8,10,17,24,56,58,65,72]
7	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[25,26,39,40,73,74,87,88]
8	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[27,28,37,38,75,76,85,86]
9	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[29,36,44,46,77,84,92,94]
10	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[30,33,43,47,78,81,91,95]
11	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[31,35,42,45,79,83,90,93]

continued ...

Table 4

No.	position	mapping
12	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[32,34,41,48,80,82,89,96]

Table 5: Wyckoff site: 12e, site symmetry: 2mm..1'

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,2,15,16,49,50,63,64]
2	$\left[-x, 0, \frac{1}{2}\right]$	[3,4,13,14,51,52,61,62]
3	$\left[\frac{1}{2}, x, 0\right]$	[5,12,20,22,53,60,68,70]
4	$\left[0, \frac{1}{2}, x\right]$	[6,9,19,23,54,57,67,71]
5	$\left[0, \frac{1}{2}, -x\right]$	[7,11,18,21,55,59,66,69]
6	$\left[\frac{1}{2}, -x, 0\right]$	[8,10,17,24,56,58,65,72]
7	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[25,26,39,40,73,74,87,88]
8	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[27,28,37,38,75,76,85,86]
9	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[29,36,44,46,77,84,92,94]
10	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[30,33,43,47,78,81,91,95]
11	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[31,35,42,45,79,83,90,93]
12	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[32,34,41,48,80,82,89,96]

Table 6: Wyckoff site: 16f, site symmetry: .3.1'

No.	position	mapping
1	$[x, x, x]$	[1,5,6,49,53,54]
2	$[x, -x, -x]$	[2,10,11,50,58,59]
3	$[-x, x, -x]$	[3,7,12,51,55,60]
4	$[-x, -x, x]$	[4,8,9,52,56,57]
5	$[-x, -x, -x]$	[13,17,18,61,65,66]
6	$[-x, x, x]$	[14,22,23,62,70,71]
7	$[x, -x, x]$	[15,19,24,63,67,72]
8	$[x, x, -x]$	[16,20,21,64,68,69]
9	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[25,29,30,73,77,78]
10	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[26,34,35,74,82,83]
11	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[27,31,36,75,79,84]
12	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[28,32,33,76,80,81]
13	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[37,41,42,85,89,90]
14	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[38,46,47,86,94,95]
15	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[39,43,48,87,91,96]
16	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[40,44,45,88,92,93]

Table 7: Wyckoff site: 24g, site symmetry: m..1'

No.	position	mapping
1	$[0, y, z]$	[1,14,49,62]
2	$[0, -y, -z]$	[2,13,50,61]

continued ...

Table 7

No.	position	mapping
3	$[0,\ y,\ -z]$	`[3,16,51,64]`
4	$[0,\ -y,\ z]$	`[4,15,52,63]`
5	$[z,\ 0,\ y]$	`[5,24,53,72]`
6	$[y,\ z,\ 0]$	`[6,21,54,69]`
7	$[-y,\ z,\ 0]$	`[7,23,55,71]`
8	$[-z,\ 0,\ y]$	`[8,22,56,70]`
9	$[-y,\ -z,\ 0]$	`[9,18,57,66]`
10	$[z,\ 0,\ -y]$	`[10,20,58,68]`
11	$[y,\ -z,\ 0]$	`[11,19,59,67]`
12	$[-z,\ 0,\ -y]$	`[12,17,60,65]`
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[25,38,73,86]`
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[26,37,74,85]`
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[27,40,75,88]`
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[28,39,76,87]`
17	$\left[z+\frac{1}{2},\ \frac{1}{2},\ y+\frac{1}{2}\right]$	`[29,48,77,96]`
18	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}\right]$	`[30,45,78,93]`
19	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ \frac{1}{2}\right]$	`[31,47,79,95]`
20	$\left[\frac{1}{2}-z,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	`[32,46,80,94]`
21	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ \frac{1}{2}\right]$	`[33,42,81,90]`
22	$\left[z+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-y\right]$	`[34,44,82,92]`
23	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}\right]$	`[35,43,83,91]`
24	$\left[\frac{1}{2}-z,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	`[36,41,84,89]`

Table 8: Wyckoff site: `48h`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,49]`
2	$[x,\ -y,\ -z]$	`[2,50]`
3	$[-x,\ y,\ -z]$	`[3,51]`
4	$[-x,\ -y,\ z]$	`[4,52]`
5	$[z,\ x,\ y]$	`[5,53]`
6	$[y,\ z,\ x]$	`[6,54]`
7	$[-y,\ z,\ -x]$	`[7,55]`
8	$[-z,\ -x,\ y]$	`[8,56]`
9	$[-y,\ -z,\ x]$	`[9,57]`
10	$[z,\ -x,\ -y]$	`[10,58]`
11	$[y,\ -z,\ -x]$	`[11,59]`
12	$[-z,\ x,\ -y]$	`[12,60]`
13	$[-x,\ -y,\ -z]$	`[13,61]`
14	$[-x,\ y,\ z]$	`[14,62]`
15	$[x,\ -y,\ z]$	`[15,63]`
16	$[x,\ y,\ -z]$	`[16,64]`
17	$[-z,\ -x,\ -y]$	`[17,65]`
18	$[-y,\ -z,\ -x]$	`[18,66]`
19	$[y,\ -z,\ x]$	`[19,67]`

continued ...

Table 8

No.	position	mapping
20	$[z,\, x,\, -y]$	[20,68]
21	$[y,\, z,\, -x]$	[21,69]
22	$[-z,\, x,\, y]$	[22,70]
23	$[-y,\, z,\, x]$	[23,71]
24	$[z,\, -x,\, y]$	[24,72]
25	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[25,73]
26	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[26,74]
27	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27,75]
28	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[28,76]
29	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[29,77]
30	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[30,78]
31	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[31,79]
32	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[32,80]
33	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[33,81]
34	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[34,82]
35	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[35,83]
36	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[36,84]
37	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[37,85]
38	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[38,86]
39	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[39,87]
40	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[40,88]
41	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[41,89]
42	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[42,90]
43	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[43,91]
44	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[44,92]
45	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[45,93]
46	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[46,94]
47	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[47,95]
48	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[48,96]